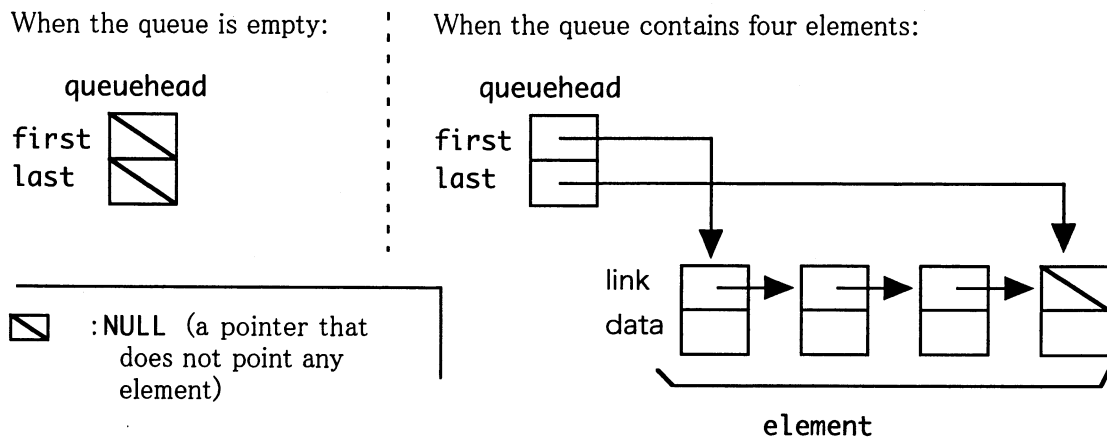


Problem I

- (1) We realize a queue that can contain integers by using pointers. The following figures show such a queue in the boxes and arrows notation.



The following is a program that adds an element to a queue written in the C language. Fill the boxes and complete the program. Note that `malloc()` is a function that allocates a memory block of a specified size (in bytes) and returns its address. In this problem, assume that the memory allocation always ends successfully. The constant `NULL` means a pointer that does not point any element.

```
#include <stdlib.h>

struct element {
    struct element *link;
    int data;
};

struct queuehead {
    struct element *first;
    struct element *last;
};
```

```
void enqueue( struct queuehead *qh,
              int data ) {
    struct element *e;
    e = malloc( sizeof(struct element) );
    e->link = (A);
    e->data = (B);
    if( qh->first == NULL ) {
        /*The queue is empty.*/
        (C)
    }
    else {
        (D)
    }
}
```

Continues on the next page.

平成19年度 筑波大学大学院 博士前期課程 システム情報工学研究科
コンピュータサイエンス専攻 入学試験問題

きそかだい すうがく じょうほうきそ すべ もんだい かいとう

基礎課題 (数学と情報基礎の全ての問題に解答すること)

じょうほう きそ
情報基礎

以下の問題I、問題IIを解答せよ。各解答は別々の答案用紙に記入すること。

Solve both Problem I and Problem II, describing in two different sheets for them.

Continues from the previous page.

- (2) A radix sort is a sorting algorithm that uses queues. In the radix sort, when a key is a nonnegative integer in base M with N digits, we use M queues as pockets (buckets). In the radix sort, we focus attention on a digit of each key from the least significant digit to the most significant digit ($n=0..N-1$). For example, when a key is decimal base with three digits, we focus attention on 1's digit, 10's digit, and 100's digit. Next, we insert each element to the corresponding pocket of the digit while we keep the order of elements with the same digit. The following is a part of a program that sorts input integers by the n -th digit with the radix sort written in the C language. This program reads data from the queue `input`, and writes the partially sorted results to the queue `output`. The function `enqueue()` is the function that adds an element to a queue, as described in Question (1). The function `dequeue()` removes an element from a queue. The function `dequeue()` returns a negative integer when the queue is empty.

```
#define M 10
struct queuehead pockets[M];

void radix_sort( struct queuehead *input,
                 struct queuehead *output, int n ) {
    int data, div, i;
    for( i=0, div = 1; i<n; i++ )
        div = div * M;
    while( (data=dequeue(input))>=0 ) {
        i = (data/div) % M;
        enqueue( &pockets[i], data );
    }
    for( i=0; i<M; i++ ) {
        while( (data=dequeue(&pockets[i]))>=0 ) {
            enqueue(output, data);
        }
    }
}
```

Consider that we give 0 to this program as the argument n , and the queue that contains the following integers in this order as the argument `input`.

52, 97, 74, 0, 37, 14, 57, 95

Answer the output of this program in the queue `output` with the same notation above. Note that the character `/` is the operator of division with any fractional part discarded, and `%` is the operator of modulo.

- (3) Consider that we give the output of Question (2) into the program, again. As the argument n , we give 1. At this time, answer the output of this program in the queue `output` with the same notation above.

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Solve both Problem I and Problem II, describing in two different sheets for them.

Problem II

- For any set A , a subset of $A \times A$ is said to be a relation over A . From (a)–(c) below, choose all the items that are true for the relation $R = \{(0, 0), (0, 1), (1, 1)\}$ over the set $B = \{0, 1\}$.
 - $(x, y) \in R \Rightarrow (y, x) \in R$.
 - $((x, y) \in R \wedge (y, z) \in R) \Rightarrow (x, z) \in R$.
 - $((x, y) \in R \wedge (y, x) \in R) \Rightarrow x = y$.
- Enumerate all relations S over the set $\{0, 1, 2\}$ such that $(x, y) \in S \Rightarrow x = y$.
- Answer the following questions in the domain of Boolean algebra. In the following, we write disjunction (logical or) of two logical expressions E_1 and E_2 as $E_1 + E_2$, conjunction (logical and) as $E_1 \cdot E_2$, true as 1, false as 0, and negation of logical expression E as $\overline{E}$, respectively. All expressions are logical expressions unless otherwise noted. Logical expressions can contain only logical variables, disjunction, conjunction, negation, 1, or 0.
 - Using a truth table, find all the pairs (p, q) such that the logical expression $(p + q) \cdot \overline{(p \cdot q)}$ is equal to 1.
 - Show a logical expression that is equal to 1 if and only if p is equal to q .
 - When we view the values of p and q as 1-digit binary numbers, show a logical expression that is equal to 1 if and only if p is greater than q .